

Sean Meehan

Secret-Level Security Clearance | April 2024-Present
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Education

Master of Science in Applied Mathematics (expected)

Tulane University

December 2026

New Orleans, LA

GPA: 3.95

- **Relevant Courses:** Data Science, Linear Models, Scientific Computing II, Probability II, Data Analysis, Algorithms, Machine Learning

Bachelor of Science in Mathematics, Minor in Computer Science

Samford University

May 2025

Birmingham, AL

GPA: 3.94

- **Relevant Courses:** Programming I & II, Data Structures and Algorithms, Programming Languages, AI and Complex Computing Methods, Statistics, Linear Algebra

Skills

- **Programming:** Python, TypeScript, MATLAB, R
- **Machine Learning:** PyTorch, TensorFlow, Keras, scikit-learn
- **Data Science:** pandas, NumPy, Matplotlib, Seaborn, SciPy, Jupyter
- **Software & Infrastructure:** Git, Docker, React, Vite, Linux, VMware, Material UI

Experience

Graduate Teaching Assistant

Tulane University

August 2025-present

New Orleans, LA

- Strengthened communication and instructional skills via weekly classes and interactions with students and faculty.
- Gained proficiency in presenting complex material to individuals with varying degrees of background and understanding

Cybersecurity & AI Research Assistant

UAH Center for Cybersecurity Research and Education

May 2023- Oct 2025

Huntsville, AL

- Developed machine learning models for Bluetooth intrusion detection using Python and modern deep learning frameworks.
- Designed automated Python/Bash pipelines to generate, clean, and preprocess cybersecurity datasets for model training.
- Engineered virtualized environments through VMware and Docker for cyberattack simulations and data collection.
- Performed EDA and statistical analysis with pandas, NumPy, and Matplotlib to identify trends and improve model design.

Projects

An ML Derived Error Correction Term for Satellite Orbital Path Propagations

- Developed an automated data acquisition pipeline using the Space-Track REST API, NASA OMNIWeb, and requests to collect satellite orbital and space weather data.
- Engineered static and time-series features from 436K+ records, SGP4 predictions, and 11 space-weather variables using pandas and NumPy; performed EDA to identify temporal and directional patterns in propagation error.
- Created a hybrid MLP/GRU neural network in PyTorch, combining static orbital parameters with sequential space-weather data while using vectorized preprocessing and batched model training to maintain computational efficiency.
- Reduced average satellite propagation error by 43.5%, improving predicted positions by approximately 3 km per observation on held-out test data.

NatTwentyOne

- Planned and developed NatTwentyOne, a customizable desktop application for managing Dungeons and Dragons characters and stories, using TypeScript, React, and Vite, comprising nearly 5,000 lines of application logic.
- Designed a layered application architecture with type-safe abstractions separating persistence (IndexedDB via Dexie), repositories, custom state management, and UI components to support maintainability and extensibility.
- Built a configurable React interface with reusable Material UI components, persistent character-management widgets, custom themes, and an interactive node-based story editor using React Flow.
- Built automated Python data pipelines using web scraping (Requests and BeautifulSoup), REST APIs, pandas, and local LLM processing to collect, clean, and generate large, structured application datasets.